

Filipi Nascimento Silva

Research Assistant Professor

Center for Science of Science & Innovation, Kellogg School of Management, Northwestern University

Updated August 2, 2026

[Website](#) [GitHub](#) [Google Scholar](#) [ORCID](#) [OpenAlex](#)

60+ Publications	2,605 Citations	23 h-index
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Profile

Research Assistant Professor at Northwestern University’s Kellogg School of Management and the Center for Science of Science & Innovation. Researcher, data scientist, and developer with expertise in dynamic complex networks, machine learning, science of science, computational social science, and interactive visualization. Creator of Helios Web and author of 60+ publications, with 2,605 citations and an h-index of 23 on Google Scholar. Ph.D. in Computational Physics with a strong interdisciplinary focus bridging physics, computer science, biology, and the social sciences.

Appointments

- Research Assistant Professor**, Center for Science of Science & Innovation, Kellogg School of Management, Northwestern University — 2025–present
- Associate Research Scientist**, Observatory on Social Media, Indiana University — 2023–2025
- Assistant Research Scientist**, Indiana University Network Science Institute — 2019–2023
- Postdoctoral Fellow**, São Carlos Institute of Physics, University of São Paulo — 2015–2019
- Visiting Research Scholar**, Indiana University Bloomington — 2017–2018

Education

- Ph.D. in Physics, Computational Physics**, São Carlos Institute of Physics, University of São Paulo — 2015
- M.Sc. in Physics, Computational Physics**, São Carlos Institute of Physics, University of São Paulo — 2009
- B.Sc. in Physics, Computational Physics**, São Carlos Institute of Physics, University of São Paulo — 2006

University of São Paulo is ranked [#1 in Latin America in the Times Higher Education Latin America University Rankings 2026](#).

Dissertation received a university-level Honorable Mention, ranking among the top three theses in the exact sciences.

Selected projects

- Scientific and engineering discovery:** Combining science-of-science evidence, structured extraction pipelines, network analysis, embedding-based search, and domain knowledge to identify promising scientific and engineering opportunities.
- Technology capability maps:** Developing embedding models that represent design choices and technical attributes across papers and patents, predict technological capabilities, and identify interpretable directions aligned with measurable progress in materials and device platforms.

- **Graph and language models of scientific impact:** Comparing random-walk embeddings, transformer representations, and graph-retrieval methods to model scientific impact, research similarity, and scholarly trajectories.
- **Mapping how science is funded:** Linking funding acknowledgments to organizations and publications to study how different forms of support shape research directions.
- **Helios Web:** Browser-native visualization and exploration of large networks and embedding spaces. [GitHub](#)

[Browse the full project portfolio.](#)

Research areas

Network science; science of science; complex systems; artificial intelligence and machine learning; computational social science; scientific and engineering discovery; graph and text embeddings; interactive visualization; community detection; information science; biological, climate, political, and urban networks.

Selected teaching and mentorship

- Instructor, **INFO-I 590: Data Visualization**, Luddy School of Informatics, Computing, and Engineering, Indiana University, Spring 2025; in person. A project-based course connecting visual perception and design to exploratory analysis, multidimensional data, maps, text and embeddings, networks, interaction, and web deployment using Python and JavaScript. [Course site](#) · [Materials](#)
- Instructor, **INFO-I 513: Usable Artificial Intelligence**, Luddy School of Informatics, Computing, and Engineering, Indiana University, Spring 2025; online. An applied introduction to data preparation, regression, classification, clustering, evaluation, feature selection, explainability, natural-language processing, embeddings, and responsible use of large language models in practical Python workflows. [Course site](#) · [Materials](#)
- Guest lecturer on complex-network visualization at the University of São Paulo.
- Graduate and undergraduate research supervision in network visualization, embedding analysis, large-language-model interfaces, and scientific software.
- Google Summer of Code mentor for Helios/FURY projects under the Python Software Foundation, 2021–2023.

Selected service

- Academic editor for PLOS ONE and PeerJ.
- Reviewer for journals and conferences spanning network science, information science, complex systems, computational social science, and physics.
- Organizer of tutorials and workshops for IC2S2, ISSI/CADRE, and the Indiana University Network Science Institute.
- Contributor to open research infrastructure including Helios Web, OSoMe tools, brainlife.io network apps, Scholarometer, and FURY.

Technical skills

Python, JavaScript, C/C++, Objective-C, WebGPU, WebGL, OpenGL, WebAssembly, parallel processing, large-scale data pipelines, network analysis, community detection, scientific visualization, natural-language processing, machine learning, graph and text embeddings, PyTorch, TensorFlow, transformer models, and LLM APIs.

Languages

Portuguese (native); English (fluent).

Profiles

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